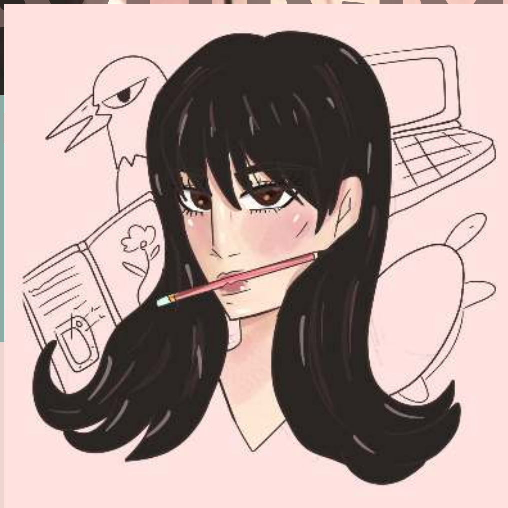

(PROBABLY ALMOST)

EVERYTHING ON THE SYLLABUS

XBI11 IAL EDEXCEL BIOLOGY

UNIT 1 & 2

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Cell Structure

Cell Structure

▼ What is a cell?

A cell is the smallest, structural, functional and fundamental unit of life.

It can exist as a part of an organism or independently.

Cells → Tissues → Organs → Organ Systems → Organism

1. Tissue: Group of similar cells/one type of cell working together for a common function.
2. Organs: Group of different tissues working together to carry out a range of different functions
3. Organ Systems: Group of organs working together as a biological system to carry out one or more functions.
4. Organism: All the organ systems working to carry out all the functions of a body

Cell Theory:

- Cells form the building blocks of living organisms
- Cells arise only by the division of existing cells
- Cells contain inherited information which controls their activities
- The cell is the functioning unit of life; metabolism (the chemical reactions of life) takes place in cells
- Given suitable conditions, cells are capable of independent existence.

▼ What are the two types of cells and what makes them fundamentally different?

1. Prokaryotic Cells - without a fully formed, enclosed nucleus. Has no membrane bound organelles.
2. Eukaryotic Cells - with a fully formed, enclosed nucleus. Has membrane bound organelles.

▼ What are the differences between Prokaryotic Cells & Eukaryotic Cells

e) State other 3 differences between eukaryote and prokaryote. [3]

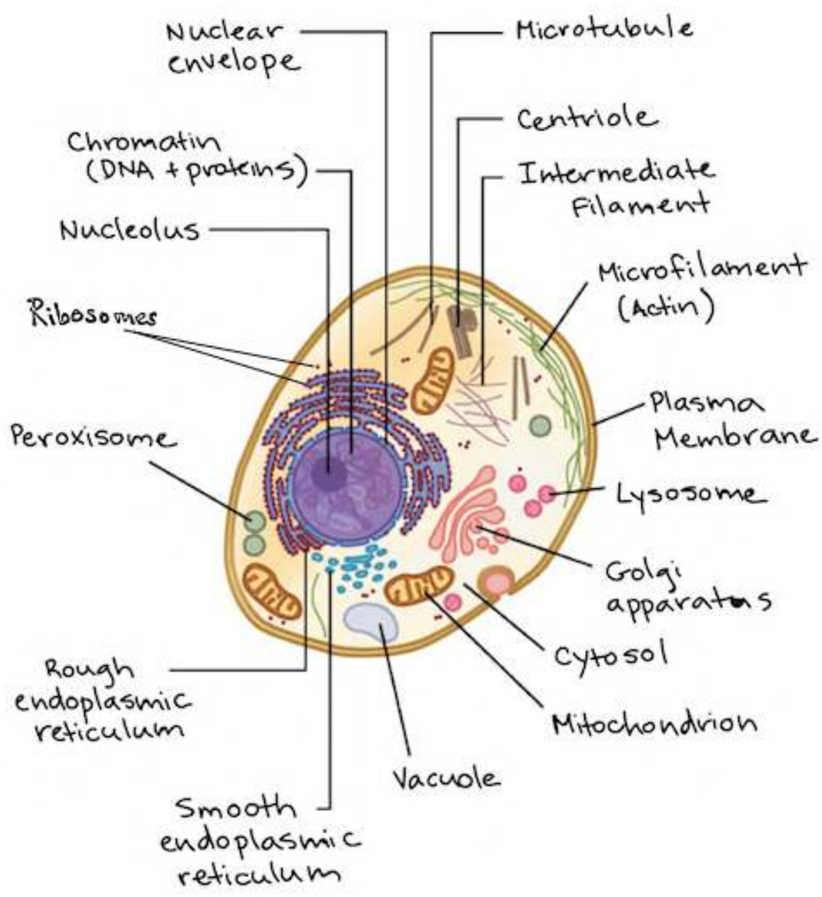
Feature	Prokaryote	Eukaryote
Size	Small (1-10 µm)	Large (10 or more µm)
Nucleus	Absent (DNA floating freely in cytoplasm)	Present (DNA contain in nucleus)
Membrane Bound Organelles	Absent	Present (mitochondria, golgi, chloroplasts, etc.)
Cell Division	Rapid process (Binary fission)	Complex process (Mitosis or meiosis)
Reproduction	Always asexual	Asexual or sexual
DNA	Circular	Linear
No of chromosomes	One--but not true chromosome	More than one
Ribosomes	70s	80s

Differences

Prokaryotic Cells	Eukaryotic Cells
Unicellular	Unicellular or Multicellular
DNA floating in the cytoplasm	DNA present in a nucleus
No nucleus or nuclear membrane	Enclosed nucleus by nuclear membrane
No membrane bound organelles	Membrane Bound Organelles present - surrounded by lipid bilayer
Simple internal structure	Complex internal structure - compartmentalization by endomembrane system
Smaller in size - about 1 - 10 um in diameter for bacteria, and 0.1 - 1 um for mycoplasmas	Larger in size - about 10 - 100 um in diameter
Has plasmids/ slime capsule/ pili	Does not have plasmids, slime capsule, pili

Prokaryotic Cells	Eukaryotic Cells
Most have a rigid cell wall, may have a capsule surrounding it, pili (for attachment) and flagella for movement	Can have
Smaller ribosomes - 70s	Larger ribosomes - 80s
Circular DNA	Linear DNA
Cell walls always present	Cells walls present in some eukaryotic cells
E.g. - Bacteria, Archaea	E.g - Animals, Plants

▼ Outline the structure of a Eukaryotic Cell



1. **The Nucleus**

    - Contains Chromatin which contains DNA, and condenses to form chromosomes which contain the genetic information of the cell.

- Controls chemical reactions within the cell by controlling the synthesis of particular enzymes.
- Responsible for protein synthesis - site of transcription
- Nucleolus - Site of ribosome synthesis

  Nuclear Envelope:

  - Double membrane/2 phospholipid bilayers that encloses the nucleus's contents and separates the nucleus from the cytoplasm.
  - Dotted with nuclear pores which permit the exchange of substances between the nucleoplasm and cytoplasm.
  - Gives the nucleus its shape.
  - Disintegrates during cellular division.

2. Ribosomes

- Site of Protein Synthesis using coded instructions from the nucleus via mRNA
- Has two types in Eukaryotic Cells - 70s and 80s

3. Endoplasmic Reticulum

Series of flattened, interconnected membrane bound sacs called cisternae. Site of lipid and protein assembly and synthesis

Smooth Type:

- No ribosomes present.
- Site of lipid and steroid synthesis
- Site of synthetic processes, detoxification of drugs.
- Site of storage
- Forms Transport Vesicles

Rough Type:

- Ribosomes present
- Site of protein synthesis.

- Modifies and processes proteins i.e. by adding carbohydrates to proteins to make glycoproteins.

4. Mitochondria:

- Diameter of 0.5 - 1 um and length of 7 um.
- Contains ribosomes & DNA
- Site of cellular respiration → produces ATP.
- Double membraned - inner membrane surrounds the matrix and is folded to form cristae
- Matrix - inner semifluid containing respiratory enzymes to break down glucose

5. Centrioles (Animal) /Centrosome (Plants)

- Releases spindle fibres during cell divison.
- Involved in flagella and cilia formation.

6. Lysosomes

- Not found in plants)
- Small organelle that produces/is filled with digestive enzymes that digests lipids, carbohydrates and proteins to be used by the cell.
- Digests waste and organelles no longer in use.

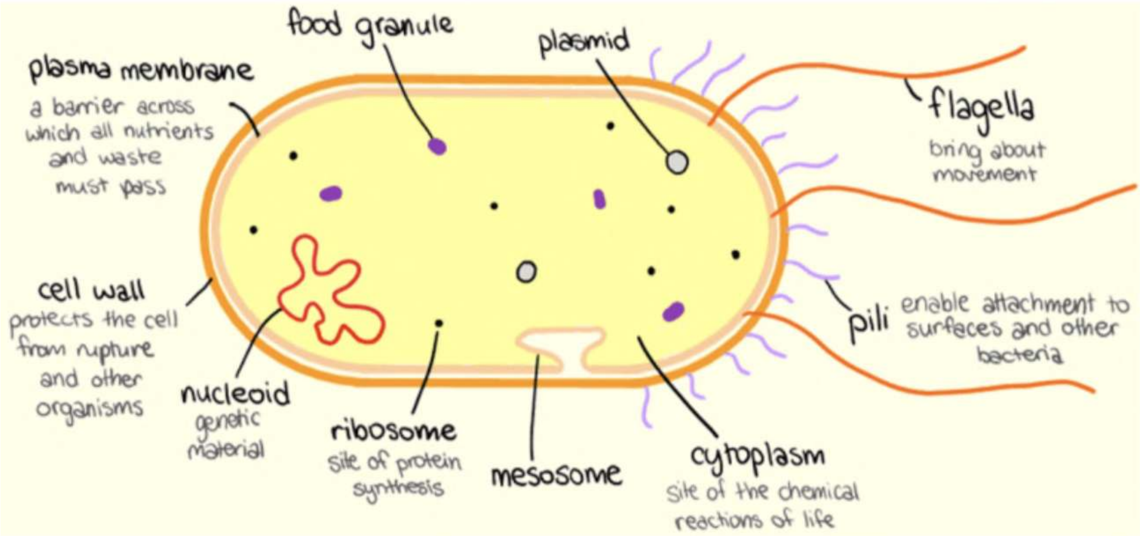
7. Golgi Apparatus

- Series of fluid filled, flattened and curved sacs (cisternae) with vesicles.
- Sorts, processes & packages proteins & lipids & other materials from ER for storage in cell or secretion outside the cell.
- Modifies proteins and enzymes, i.e. by the addition of carbohydrates to form glycoprotein.
- Then it packages the proteins and enzymes.
- Also produces secretory vesicles and lysosomes.

8. Vacuoles

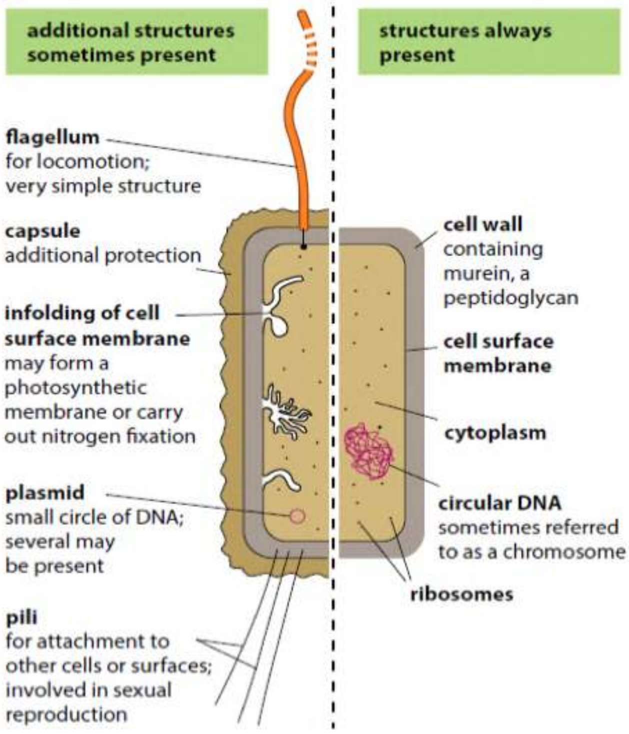
- Sac-like structures that store materials i.e. water, salts, proteins, carbohydrates, etc.

▼ Outline the structure of a Prokaryotic Cell



1. *Cell wall* – rigid outer covering; maintains shape and prevents bursting (lysis)
2. *Slime capsule* – a thick polysaccharide layer used for protection against dessication (drying out) and phagocytosis
3. **Cell membrane* – Semi-permeable and selective barrier surrounding the cell which allows the movement of molecules in an out of the cell.
4. **Cytoplasm* – internal fluid component of the cell containing essential materials i.e. proteins, water, etc. Site of metabolic reactions
5. **Nucleoid* – region of the cytoplasm where the DNA is located (DNA strand is circular and called a genophore)
6. *Plasmids* – autonomous circular DNA molecules that may be transferred between bacteria (horizontal gene transfer). The DNA molecule capable of replication.
7. *Ribosomes* – complexes of RNA and protein that are responsible for polypeptide synthesis (prokaryote ribosome = 70S)

8. *Mesosomes* - Infoldings of the inner membrane containg enzymes for respiration.
9. *Flagella* – Long, slender projections containing a motor protein that enables movement (singular: flagellum) → gives the cell motility and may aid in sexual conjugation.
10. *Pili* – Hair-like extensions that enable adherence to surfaces (attachment pili) or mediate bacterial/sexual conjugation (sex pili)



▼ Compare rER and sER

- Both consist of membrane bound sacs called cisternae that are interconnected.
- However, sER has 80s ribosomes, sER does not.

Protein Transport

▼ How do organelles form extracellular enzymes?

- Extracellular enzymes are enzymes secreted by cells into their external environment.

- These are proteins that are synthesised in ribosomes free in the cytoplasm or attached to the RER.
- After initial synthesis, the polypeptide is sent to the Golgi apparatus, where it is folded into its 3D shape and components are added for specific functions so eg. Glycoproteins, and then packaged in a Golgi vesicle which can then fuse with the cell surface membrane to release the enzymes outside of the cell via exocytosis.
    1. Translation occurs on the ribosomes and enzymes which are synthesised are transported to the rER, where it is folded into its 3D shape/tertiary structure.
    2. Enzymes are then packaged into transport vesicles by the rER to move to the Golgi apparatus where the golgi apparatus modifies the enzyme i.e. by adding carbohydrates to proteins to form glycoproteins.
    3. Golgi apparatus then packages the enzyme in to secretory vesicles which transports the enzyme to the cell surface membrane, and the vesicle fuses with the cell membrane and releases the enzyme out of the cell (exocytosis).
- These vesicles are important to ensuring the enzymes only catalyse reactions, such as the breakdown of biological molecules, where required, in this case outside of the cell.

Question Number	Acceptable Answers	Additional Guidance	Mark
***5(b)**	(QWC – Spelling of technical terms *(shown in italics)* must be correct and the answer must be organised in a logical sequence) 1. idea that the *enzyme* is transported (through the *cell / cytoplasm*) in the rER ; 2. idea that in the rER *enzyme* is folded; 3. idea of *enzyme* being packaged into (transport) *vesicles* (by the rER) to { move to / fuse with / eq } the *Golgi apparatus* ; 4. credit description of modification ; 5. idea of *enzyme* being transferred in (*secretory*) *vesicles* from the *Golgi apparatus* to the cell (surface) *membrane* ; 6. *vesicles* (containing *enzyme*) fuse with cell (surface) *membrane / exocytosis* ;	QWC emphasis – spelling **2** e.g. forms {3-D shape, *secondary / tertiary* structure } **4 ACCEPT** e.g. addition / removal of *sugars, glycosides, carbohydrate*, or activation of *enzyme* **5. IGNORE** lysosomes	(5)

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